BCDPPU

2008 年 9 月 19 日

第 1 章

論理譜

```
{ ============================================================== }
{    BCD 変換器                                              }
{ ============================================================== }
logicname BCD

library ppu   { PPU をライブラリに読み込み }

{ ============================================================== }
{    実効譜                                                }
{ ============================================================== }
entity main
{ -------------------------------------------------------------- }
{    入出力                                                }
{ -------------------------------------------------------------- }
input  RESET;
input  BDATA[8];
input  SCLK;
input  CONV;
output LED7SEG[7];
output EOC;
output KETA[3];

{ -------------------------------------------------------------- }
{    PPU 1 引用                                            }
{ -------------------------------------------------------------- }
bitn   DATAIN[32];
bitn   PORT0IN[8];
bitn   PORT1IN[8];
bitn   int_ppu[49];
bitn   ADDRESS[8];
bitn   PORT0ADD[8];
bitn   PORT1ADD[8];
bitn   PORT0OUT[8];
bitn   RW;
bitn   INT;

{ -------------------------------------------------------------- }
{    PPU 1 内部信号                                        }
{ -------------------------------------------------------------- }
bitn   reg0R[8];
bitn   reg1R[8];
bitr   reg2R[8];
bitr   reg3R[8];
bitr   reg4R[8];
bitr   reg5R[8];
bitn   reg6R;
bitr   reg7R;
bitn   flgE;
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```
bitr   flgZ;
bitn   divout[8];
bitr   divina[8];
bitr   divinb[8];
bitn   code23p;
bitn   code19p;
bitn   code24p;
bitn   code27p;
bitn   opout[8];
bitn   opina[8];
bitn   opinb[8];
bitn   datain[32];

{ ------------------------------------------------------------- }
{    PPU 2 引用                                              }
{ ------------------------------------------------------------- }
bitn   DATAINb[32];
bitn   PORT0INb[8];
bitn   PORT1INb[8];
bitn   intb_ppu[49];
bitn   ADDRESSb[8];
bitn   PORT0ADDb[8];
bitn   PORT1ADDb[8];
bitn   PORT0OUTb[8];
bitn   RWb;
bitn   INTb;
bitn   datainb[32];

{ ------------------------------------------------------------- }
{    PPU 2 内部信号                                          }
{ ------------------------------------------------------------- }
bitn   reg0Rb;
bitr   reg1Rb[2];
bitr   reg2Rb[3];
bitr   reg3Rb[4];
bitr   reg4Rb[7];
bitr   reg5Rb[4];
bitr   reg6Rb[4];
bitr   reg7Rb[4];
bitr   flgZb;
bitn   codeb21p;
bitn   codeb19p;
bitn   opbout[8];
bitn   opbina[8];
bitn   opbinb[8];

{ ------------------------------------------------------------- }
{    検査端子                                                }
{ ------------------------------------------------------------- }

output T0P[8];  T0P=PORT0ADD;
output T1P;     T1P=RW;
output T2P[8];  T2P=ADDRESS;
output T3P[32]; T3P=DATAIN;
output T4P[8];  T4P=PORT0IN;
output T5P[8];  T5P=PORT1IN;
output T6P[8];  T6P=PORT1ADD;
output T7P[8];  T7P=PORT0OUT;
output T8P[8];  T8P=reg0R;
output T9P[8];  T9P=reg1R;
output T10P[8]; T10P=divina;
output T11P[8]; T11P=divinb;
output T12P[8]; T12P=reg2R;
output T13P[8]; T13P=reg3R;
output T14P;    T14P=flgZ;
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```
output T15P[8]; T15P=reg4R;
output T16P[8]; T16P=reg5R;
output T17P[2]; T17P=reg1Rb;
output T20P[8];  T20P=PORT0ADDb;
output T21P;     T21P=RWb;
output T22P[8];  T22P=ADDRESSb;
output T23P[32]; T23P=DATAINb;
output T24P[8];  T24P=PORT0INb;
output T25P[8];  T25P=PORT1INb;
output T26P[8];  T26P=PORT1ADDb;
output T27P[8];  T27P=PORT0OUTb;
output T28P;     T28P=flgZb;
output T29P[3];  T29P=reg2Rb;
output T30P[4];  T30P=reg3Rb;
output T31P[7];  T31P=reg4Rb;
output T32P[4];  T32P=reg5Rb;
output T33P[4];  T33P=reg6Rb;
output T34P[4];  T34P=reg7Rb;
output T35P;     T35P=code23p;
output T36P;     T36P=divout.8;
output T37P;     T37P=flgE;
output T38P[8];  T38P=opout;
output T39P[8];  T39P=opina;
output T40P[8];  T40P=opinb;
output T41P;     T41P=code19p;
output T42P;     T42P=code24p;
output T43P[8];  T43P=opbout;
output T44P[8];  T44P=opbina;
output T45P[8];  T45P=opbinb;
output T46P;     T46P=codeb19p;
output T47P;     T47P=codeb21p;

{ --------------------------------------------------------------- }
{    出力代入                                              }
{ --------------------------------------------------------------- }
   LED7SEG=reg4Rb;
   EOC=reg7R;
   KETA=reg2Rb;

{ --------------------------------------------------------------- }
{    PPU 1 導入                                              }
{ --------------------------------------------------------------- }
   int_ppu=ppu(RESET,DATAIN,PORT0IN,PORT1IN,INT);

   int_ppu.49=1;

   ADDRESS=int_ppu.0:7;
   PORT0ADD=int_ppu.16:23;
   PORT1ADD=int_ppu.24:31;
   PORT0OUT=int_ppu.40:47;
   RW=int_ppu.48;

{ --------------------------------------------------------------- }
{    PPU 1 プログラム導入                                    }
{ --------------------------------------------------------------- }
   preparation codegendw(ADDRESS,datain,bcda)

   DATAIN.0:7  =datain.24:31;
   DATAIN.8:15 =datain.16:23;
   DATAIN.16:23=datain.8:15;
   DATAIN.24:31=datain.0:7;

{ --------------------------------------------------------------- }
{    PPU 1 命令解読                                          }
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```
{ ---------------------------------------------------------------- }
   switch(DATAIN.24:31)
      case 19: code19p=1;
      case 23: code23p=1;
      case 24: code24p=1;
      case 27: code27p=1;
   endswitch

{ ---------------------------------------------------------------- }
{    PPU 1 除算器                                     }
{ ---------------------------------------------------------------- }
   if (!code23p)
      divout=divina/divinb;
   endif

{ ---------------------------------------------------------------- }
{    PPU 1 除算器入力 a                               }
{ ---------------------------------------------------------------- }
   if (code23p)
      switch(PORT0ADD)
         case 0:  divina=PORT0IN;
         default: divina=divina;
      endswitch
   else
      divina=divina;
   endif

{ ---------------------------------------------------------------- }
{    PPU 1 除算器入力 b                               }
{ ---------------------------------------------------------------- }
   if (code23p)
      divinb=DATAIN.8:15;
   else
      divinb=divinb;
   endif

{ ---------------------------------------------------------------- }
{    PPU 1 レジスタ代入                               }
{ ---------------------------------------------------------------- }
   reg0R=BDATA;
   reg1R=divout;
   reg6R=CONV;

{ ---------------------------------------------------------------- }
{    PPU 1 レジスタ R2                                }
{ ---------------------------------------------------------------- }
   if (RESET)
      reg2R=0;
   else
      if (RW)
         switch(PORT0ADD)
            case 2:  reg2R=PORT0OUT;
            default: reg2R=reg2R;
         endswitch
      else
         reg2R=reg2R;
      endif
   endif

{ ---------------------------------------------------------------- }
{    PPU 1 レジスタ R3                                }
{ ---------------------------------------------------------------- }
   if (RESET)
      reg3R=0;
   else
```

```
    if (code27p)
       switch(PORTOADD)
          case 3:  reg3R=reg3R>>DATAIN.8:11;
          default: reg3R=reg3R;
       endswitch
    else
       if (RW)
          switch(PORTOADD)
             case 3:  reg3R=PORTOOUT;
             default: reg3R=reg3R;
          endswitch
       else
          reg3R=reg3R;
       endif
    endif
 endif

{ ---------------------------------------------------------------- }
{    PPU 1 レジスタ R4                                 }
{ ---------------------------------------------------------------- }
 if (RESET)
    reg4R=0;
 else
    if (code24p)
       switch(PORTOADD)
          case 4:  reg4R=opout;    { SUB r4,n }
          default: reg4R=reg4R;
       endswitch
    else
       if (RW)
          switch(PORTOADD)
             case 4:  reg4R=PORTOOUT;
             default: reg4R=reg4R;
          endswitch
       else
          reg4R=reg4R;
       endif
    endif
 endif

{ ---------------------------------------------------------------- }
{    PPU 1 レジスタ R5                                 }
{ ---------------------------------------------------------------- }
 if (RESET)
    reg5R=0;
 else
    if (code24p)
       switch(PORTOADD)
          case 5:  reg5R=opout;    { SUB r5,n }
          default: reg5R=reg5R;
       endswitch
    else
       if (RW)
          switch(PORTOADD)
             case 5:  reg5R=PORTOOUT;
             default: reg5R=reg5R;
          endswitch
       else
          reg5R=reg5R;
      endif
    endif
 endif

{ ---------------------------------------------------------------- }
{    PPU 1 レジスタ R7                                 }
```

```
{ -------------------------------------------------------------- }
   if (RESET)
      reg7R=0;
   else
      if (RW)
         switch(PORT0ADD)
            case 7:  reg7R=PORT0OUT.0;
            default: reg7R=reg7R;
         endswitch
      else
         reg7R=reg7R;
      endif
   endif

{ -------------------------------------------------------------- }
{   PPU 1 レジスタ FE                               }
{ -------------------------------------------------------------- }
   flgE=divout.8;

{ -------------------------------------------------------------- }
{   PPU 1 演算器                                    }
{ -------------------------------------------------------------- }
   if (code19p) opout.0=opina==opinb;  endif
   if (code24p) opout=opina-opinb; endif

{ -------------------------------------------------------------- }
{   PPU 1 演算器入力 a                              }
{ -------------------------------------------------------------- }
   switch(PORT0ADD)
      case 1: opina=reg1R;
      case 2: opina=reg2R;
      case 4: opina=reg4R;
      case 5: opina=reg5R;
   endswitch

{ -------------------------------------------------------------- }
{   PPU 1 演算器入力 b                              }
{ -------------------------------------------------------------- }
   if (code24p)
      switch(PORT1ADD)
         case 3: opinb=reg3R;
      endswitch
   endif

   if (code19p) opinb=DATAIN.8:15; endif

{ -------------------------------------------------------------- }
{   PPU 1 レジスタ FZ                               }
{ -------------------------------------------------------------- }
   if (RESET)
      flgZ=0;
   else
      if (code19p)
         switch(DATAIN.16:23)
            case 1:  flgZ=opout.0;   { CP r1,n }
            case 2:  flgZ=opout.0;   { CP r2,n }
            default: flgZ=flgZ;
         endswitch
      else
         flgZ=flgZ;
      endif
   endif

{ -------------------------------------------------------------- }
{   PPU 1 レジスタ枠 0 入力                         }
```

```
{ -------------------------------------------------------------- }
      switch(PORT0ADD)
         case 0:  PORT0IN=reg0R;
         case 1:  PORT0IN=reg1R;
         case 2:  PORT0IN=reg2R;
         case 3:  PORT0IN=reg3R;
         case 4:  PORT0IN=reg4R;
         case 5:  PORT0IN=reg5R;
         case 6:  PORT0IN.0=reg6R;
         case 7:  PORT0IN.0=reg7R;
         case 17: PORT0IN.0=flgZ;
         case 18: PORT0IN.0=flgE;
      endswitch

{ -------------------------------------------------------------- }
{   PPU 1 レジスタ枠 1 入力                         }
{ -------------------------------------------------------------- }
      switch(PORT1ADD)
         case 0:  PORT1IN=reg0R;
         case 1:  PORT1IN=reg1R;
         case 2:  PORT1IN=reg2R;
         case 3:  PORT1IN=reg3R;
         case 4:  PORT1IN=reg4R;
         case 5:  PORT1IN=reg5R;
         case 6:  PORT1IN.0=reg6R;
         case 7:  PORT1IN.0=reg7R;
         case 17: PORT1IN.0=flgZ;
         case 18: PORT1IN.0=flgE;
      endswitch

{ -------------------------------------------------------------- }
{   PPU 2 導入                                      }
{ -------------------------------------------------------------- }
   intb_ppu=ppu(RESET,DATAINb,PORT0INb,PORT1INb,INTb);

   intb_ppu.49=1;

   ADDRESSb=intb_ppu.0:7;
   PORT0ADDb=intb_ppu.16:23;
   PORT1ADDb=intb_ppu.24:31;
   PORT0OUTb=intb_ppu.40:47;
   RWb=intb_ppu.48;

{ -------------------------------------------------------------- }
{   PPU 2 プログラム導入                            }
{ -------------------------------------------------------------- }
   preparation codegendw(ADDRESSb,datainb,bcdb)

   DATAINb.0:7  =datainb.24:31;
   DATAINb.8:15 =datainb.16:23;
   DATAINb.16:23=datainb.8:15;
   DATAINb.24:31=datainb.0:7;

{ -------------------------------------------------------------- }
{   PPU 2 命令解読                                  }
{ -------------------------------------------------------------- }
   switch(DATAINb.24:31)
      case 21: codeb21p=1;   { ADD rx,n }
      case 19: codeb19p=1;   { CP rx,n }
   endswitch

{ -------------------------------------------------------------- }
{   PPU 2 レジスタ FZ                               }
{ -------------------------------------------------------------- }
   if (RESET)
```

```
      flgZb=0;
   else
      if (codeb19p)
         switch(PORT0ADDb)
            case 1:  flgZb=opbout.0;    { CP r1,n }
            case 3:  flgZb=opbout.0;    { CP r3,n }
            default: flgZb=flgZb;
         endswitch
      else
         flgZb=flgZb;
      endif
   endif

{ ----------------------------------------------------------------- }
{   PPU 2 演算器                                          }
{ ----------------------------------------------------------------- }
   if (codeb19p) opbout.0=opbina==opbinb; endif
   if (codeb21p) opbout=opbina+opbinb; endif

{ ----------------------------------------------------------------- }
{   PPU 2 演算器入力 a                                     }
{ ----------------------------------------------------------------- }
   switch(PORT0ADDb)
      case 1: opbina.0:1=reg1Rb;
      case 3: opbina.0:3=reg3Rb;
   endswitch

{ ----------------------------------------------------------------- }
{   PPU 2 演算器入力 b                                     }
{ ----------------------------------------------------------------- }
   if (codeb19p|codeb21p) opbinb=DATAINb.8:15; endif

{ ----------------------------------------------------------------- }
{   PPU 2 レジスタ R0 R5 R6 R7                              }
{ ----------------------------------------------------------------- }
   reg0Rb=SCLK;

   if (RESET)
      reg5Rb=0;
      reg6Rb=0;
      reg7Rb=0;
   else
      if (reg7R)
         reg5Rb=reg5R.0:3;
         reg6Rb=reg4R.0:3;
         reg7Rb=reg2R.0:3;
      else
         reg5Rb=reg5Rb;
         reg6Rb=reg6Rb;
         reg7Rb=reg7Rb;
      endif
   endif

{ ----------------------------------------------------------------- }
{   PPU 2 レジスタ R1                                       }
{ ----------------------------------------------------------------- }
   if (RESET)
      reg1Rb=0;
   else
      if (codeb21p)              ADD r1,n
         reg1Rb=opbout.0:1;
      else
         if (RWb)
            switch(PORT0ADDb)
               case 1:  reg1Rb=PORT0OUTb.0:1;
```

```
            default: reg1Rb=reg1Rb;
         endswitch
      else
         reg1Rb=reg1Rb;
      endif
   endif
endif

{ ---------------------------------------------------------------- }
{    PPU 2 レジスタ R2                                   }
{ ---------------------------------------------------------------- }
   if (RESET)
      reg2Rb=0;
   else
         if (RWb)
            switch(PORTOADDb)
               case 2:  reg2Rb=PORTOOUTb.0:2;
               default: reg2Rb=reg2Rb;
            endswitch
         else
            reg2Rb=reg2Rb;
         endif
   endif

{ ---------------------------------------------------------------- }
{    PPU 2 レジスタ R3                                   }
{ ---------------------------------------------------------------- }
   if (RESET)
      reg3Rb=0;
   else
         if (RWb)
            switch(PORTOADDb)
               case 3:  reg3Rb=PORTOOUTb.0:3;
               default: reg3Rb=reg3Rb;
            endswitch
         else
            reg3Rb=reg3Rb;
         endif
   endif

{ ---------------------------------------------------------------- }
{    PPU 2 レジスタ R4                                   }
{ ---------------------------------------------------------------- }
   if (RESET)
      reg4Rb=0;
   else
         if (RWb)
            switch(PORTOADDb)
               case 4:  reg4Rb=PORTOOUTb.0:6;
               default: reg4Rb=reg4Rb;
            endswitch
         else
            reg4Rb=reg4Rb;
         endif
   endif

{ ---------------------------------------------------------------- }
{    PPU 2 1 番目レジスタ入力                              }
{ ---------------------------------------------------------------- }
      switch(PORTOADDb)
         case 0:  PORTOINb.0=reg0Rb;
         case 1:  PORTOINb.0:1=reg1Rb;
         case 2:  PORTOINb.0:2=reg2Rb;
         case 3:  PORTOINb.0:3=reg3Rb;
         case 4:  PORTOINb.0:6=reg4Rb;
```

```
          case 5:  PORT0INb.0:3=reg5Rb;
          case 6:  PORT0INb.0:3=reg6Rb;
          case 7:  PORT0INb.0:3=reg7Rb;
          case 17: PORT0INb.0=flgZb;
       endswitch

{ --------------------------------------------------------------- }
{    PPU 2 2番目レジスタ入力                                      }
{ --------------------------------------------------------------- }
       switch(PORT1ADDb)
          case 0:  PORT1INb.0=reg0Rb;
          case 1:  PORT1INb.0:1=reg1Rb;
          case 2:  PORT1INb.0:2=reg2Rb;
          case 3:  PORT1INb.0:3=reg3Rb;
          case 4:  PORT1INb.0:6=reg4Rb;
          case 5:  PORT1INb.0:3=reg5Rb;
          case 6:  PORT1INb.0:3=reg6Rb;
          case 7:  PORT1INb.0:3=reg7Rb;
          case 17: PORT1INb.0=flgZb;
       endswitch

ende
```

```
{ ================================================================ }
{    機能実行譜                                                    }
{ ================================================================ }
entity sim
output RESET;
output BDATA[8];
output SCLK;
output CONV;
output LED7SEG[7];
output EOC;
output KETA[3];

input  simres;

bitr   tc[10];

   simres=0;
   if (!simres) tc=tc+1; endif

   part main(RESET,BDATA,SCLK,CONV,LED7SEG,EOC,KETA)

   if (tc<5) RESET=1; endif

   BDATA=234;

   SCLK=tc.5;

   if ((tc>200)&(tc<210)) CONV=1; endif

ende

endlogic
```